Bipartite graphs of diameter 5 from generalized quadrangles: level graphs, surgery and hubs

Draft of 26 October 2026, written for Ignacio López and Gabriela Araujo-Pardo by the degree–diameter lab (github.com/dogaozden/degree-diameter; Claude Code sessions coordinated by Doğa Özden). Job "thm-gq-d5".

This note does for diameter 5 and generalized quadrangles what research/theorems/D7_HEXAGON.md (branch claude/d7-hexagon-theorems-4w1prt) does for diameter 7 and the split Cayley hexagon: $2r+1$ copies of the incidence graph of a generalized quadrangle of order (q,q), joined by perfect matchings, have diameter 5; deleting some matching edges and attaching new "hub" vertices to the freed ends adds vertices. We state exactly which deletions survive, prove it, and give the resulting families and one new record, $(13,5)$.

Every claim is labelled **Theorem** (with a complete proof here), **Computer-checked** (a computation, with the program and its output in the repository), or **Heuristic** (an estimate, no proof). The neighbourhood of every vertex of every graph is given explicitly (Definitions 2.1 and 3.1, and for the record the solution file of §11).

File paths refer to the repository github.com/dogaozden/degree-diameter, branch claude/gq-diameter-5-xa5d3d. The tools are in research/theorems/d5gq/; the Haiku squad rounds that searched for sets and ran small-q tests are in research/squads/gqd5-r1/ and gqd5-r2/ (log: research/squads/LOG.md).

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0. Results

Notation: $n_b(\Delta, D)$ is the largest order of a bipartite graph with maximum degree $\leq \Delta$ and diameter $\leq D$. For a prime power q put

$$N = N(q) = (q+1)(q^2+1),$$

the number of points (and of lines) of a generalized quadrangle of order (q, q). The level graph of §2 with $2r+1$ levels has order $(2r+1) \cdot 2N(q)$ and degree $q+1+r$. In the 2013 table [MS13, Table 4] the cells $(10,5)$–$(15,5)$ have exactly this value for some (q, r): $(10,5) = 5 \cdot 2N(7)$, $(11,5) = 5 \cdot 2N(8)$, $(12,5) = 5 \cdot 2N(9)$, $(13,5) = 7 \cdot 2N(9)$, $(14,5) = 9 \cdot 2N(9)$, $(15,5) = 7 \cdot 2N(11)$ (§16 on their origin).

New record. $n_b(13,5) \geq 11{,}636$ (table 11,480; the lab's previous value 11,486). The graph: seven levels over $Q(4,9)$ and its dual $W(9)$, an ovoid freed on five transitions, a computer-found set of 66 lines with 30 concurrent pairs freed on five more, two-point local dualities on the other eleven, $86 + 70$ hubs (§11). It was checked by a breadth-first search from every vertex (§15).

Theorems.

- **Theorem A** (§2; these graphs have the orders of the table values): any typing, any bijections: diameter ≤ 5.
- **Theorems 3.5, 3.6** (assembly): if every transition is *repaired* (or, more generally, every two levels are within distance 5 of each other) and same-colour hubs are within distance 4, then $n_b(q+1+r, 5) \geq (2r+1) \cdot 2N + K_{\mathcal{A}} + K_{\mathcal{B}}$.
- **Theorem E** (§4): a transition with matching M and freed set R is repaired **if and only if** two covering conditions (E2), (E3) hold. This is new compared with the $D = 7$ note, where only sufficient conditions were given.
- **Theorem B** (§5): a transition whose matching extends to an automorphism (the identity between a quadrangle and its dual, or a duality) is repaired if and only if its freed set is a **partial ovoid** (resp. partial spread). The "segment sets" of the hexagon do not survive at $D = 5$.
- **Theorem C** (§6): a local duality on a non-collinear freed set repairs a transition; at $D = 5$ the condition on $M(R)$ is not needed. **Corollary 6.2**: for q odd, a local duality between $Q(4,q)$ and itself (or $W(q)$ and itself) exists on at most **two** points, because point triads of $Q(4,q)$ have 0 or 2 centres and line triads 1 or $q+1$. **Proposition 6.4**: two points always work.
- **Proposition 7.2** (§7): freed lines may come in concurrent pairs if the $\mathcal{A}$-hub at the slot of each line sits at a point of its partner (e.g. the ovoid point).
- **Families** (§10, Corollaries 10.1–10.4), valid for every q, with explicit sets and explicit hub graphs. For example, for q odd, $q \geq 5$: $n_b(q+4, 5) \geq 14N(q) + K_{\mathcal{A}} + K_{\mathcal{B}}$, where the explicit sets allow up to $\min(q^2+5,\ 6q)$ and $3q+3$ hubs and the explicit hub graphs of Lemma 9.1 decide how many are used. With those hub graphs the families exceed the 2013 table in the cells $(10,5)$–$(15,5)$ (§14).

What does not work (§§12–14): the level-graph family stays below the lab's values (doubled-matching lifts and route-chart lifts) in $(10,5)$, $(11,5)$, $(12,5)$, $(14,5)$, $(15,5)$, $(16,5)$, and far below at $\Delta \leq 9$. At $(15,5)$ the best graph of this kind we built has 20,713 vertices (relaxed set of 87 lines of $Q(4,11)$), 23 short of the lab's 20,736; §14 explains which quantity limits it.

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1. Generalized quadrangles

1.1 Definition and distances

A *generalized quadrangle* (GQ) of order (s,t), $s,t \geq 1$, is an incidence structure $\mathcal{S} = (P, L, \mathrm{I})$ of points and lines in which every line has $s+1$ points, every point is on $t+1$ lines, two points are on at most one common line, and

(GQ) for every point p and line ℓ with $p \notin \ell$ there is exactly one point of ℓ collinear with p (equivalently, exactly one line through p meets ℓ).

Then $|P| = (s+1)(st+1)$ and $|L| = (t+1)(st+1)$ [PT, 1.2.1]. The *dual* $\mathcal{S}^D$ (points and lines exchanged) is a GQ of order (t,s). A *duality* of $\mathcal{S}$ is an isomorphism $\mathcal{S} \to \mathcal{S}^D$.

$I = I(\mathcal{S})$ is the incidence graph (vertex set $P \cup L$). $d(x,y)$ is the distance in I, $\Gamma_i(x) = \{y : d(x,y) = i\}$. Points are at even distance from points; $\Gamma_2(p)$ is the set of the $s(t+1)$ points collinear with p, and $\Gamma_2(\ell)$ the set of the $t(s+1)$ lines meeting ℓ. For a set X of points, $X^\perp$ is the set of points collinear with (or equal to) every point of X; $\{x,y\}^\perp$ for non-collinear x, y has $t+1$ points, and its points are the *centres* of a triad $\{x,y,z\}$ when they are also collinear with z.

We use:

(F1) I is connected, bipartite, of diameter 4 and girth 8; points have degree $t+1$, lines $s+1$. *Proof.* Two points are at distance 2 (collinear) or, by (GQ) applied to p and any line through p', at distance 4; a point and a line are at distance 1 or 3 by (GQ); two lines are at distance ≤ 4 dually. A cycle of length 4 would be two points on two lines; a 6-cycle is a triangle, excluded by (GQ). □

(F2) If $d(x,y) \leq 3$ there is a unique geodesic from x to y.

(F3) Covering lemma. For elements x, y there is $z \in \Gamma_2(x)$ with $d(z,y) \leq 1$ if x, y have different types and $d(z,y) \leq 2$ if they have the same type. Moreover, if x and y have different types, some element of $\Gamma_2(x)$ is incident with y. *Proof.* If $d(x,y) \geq 2$ take z at distance 2 from x on a geodesic. If $d(x,y) = 1$ take $z \in \Gamma_1(y) \setminus \{x\}$ (it exists as $s, t \geq 1$); then $z \in \Gamma_2(x)$ and z is incident with y. If $x = y$ any $z \in \Gamma_2(x)$ has $d(z,y) = 2$. For different types $d(x,y) \in \{1,3\}$, and in both cases the z found is incident with y. □

(F4) A line ℓ not through p contains exactly one point of $\Gamma_2(p)$; every line through p contains s points of $\Gamma_2(p)$. In particular $\Gamma_2(p)$ meets every line. (This is (GQ).)

1.2 The classical quadrangles in coordinates

q is a prime power. All our computations use the following coordinates (research/theorems/d5gq/gqlib.py).

$W(q)$, order (q,q): the points of $\mathrm{PG}(3,q)$, and the lines of $\mathrm{PG}(3,q)$ that are totally isotropic for the symplectic form

$$w(x,y) = x_0y_2 + x_1y_3 - x_2y_0 - x_3y_1.$$

Two distinct points x, y are collinear in $W(q)$ if and only if $w(x,y) = 0$. The $q+1$ lines through x are the lines through x in the plane $x^{\perp_w}$.

$Q(4,q)$, order (q,q): the points of the parabolic quadric

$$X_0X_1 + X_2^2 + X_3X_4 = 0 \quad \text{in } \mathrm{PG}(4,q),$$

and the lines on it. We use it through the Klein correspondence: a line $\langle x, y\rangle$ of $\mathrm{PG}(3,q)$ has Plücker coordinates $p_{ij} = x_iy_j - x_jy_i$ on the Klein quadric $p_{01}p_{23} - p_{02}p_{13} + p_{03}p_{12} = 0$; it is totally isotropic

for w exactly when $p_{02}+p_{13}=0$, and then $(X_0,\dots,X_4)=(p_{01},p_{23},p_{02},p_{03},p_{12})$ is a point of the quadric above. This maps the lines of $W(q)$ bijectively onto the points of $Q(4,q)$, and the pencils of $W(q)$ onto the lines of $Q(4,q)$; so $Q(4,q)\cong W(q)^D$ [PT, 3.2.1]. In the programs $Q(4,q)$ is literally `W(q).dual()`: point i of $Q(4,q)$ is line i of $W(q)$ and line j of $Q(4,q)$ is point j of $W(q)$.

$Q(5,q)$, order (q,q^2): the elliptic quadric $X_0X_1+X_2X_3+f(X_4,X_5)=0$ in $\mathrm{PG}(5,q)$, f an irreducible binary quadratic form; $H(3,q^2)=Q(5,q)^D$, order (q^2,q). $T_2(O)$ for an oval O of $\mathrm{PG}(2,q)$ (order (q,q); for q odd O is a conic and $T_2(O)\cong Q(4,q)$). $AS(q)$, q odd, order $(q-1,q+1)$; we build it as the Payne derivative $P(W(q),x)$ at a point x (points: the q^3 points not collinear with x; lines: the lines of $W(q)$ not through x, restricted, and the hyperbolic lines $\{x,y\}^{\perp\perp}$ minus x) [PT, 3.2.6]. Only $W(q)$ and $Q(4,q)$ enter the theorems of §§2–11; §13 explains why.

All generators were checked against the GQ axioms by computer for $q\le 5$ (W, $Q(4,q)$), $q\le 3$ ($Q(5,q)$, $H(3,q^2)$) and $q=3,5$ ($AS(q)$) (research/theorems/d5gq/gqlib.py, self-test).

1.3 Ovoids, spreads, triads

A *partial ovoid* is a set of pairwise non-collinear points (in a GQ "non-collinear" and "opposite" coincide, as the diameter of I is 4); an *ovoid* is a partial ovoid meeting every line, i.e. with $st+1$ points. Dually *partial spread* and *spread*. In $W(q)$ and $Q(4,q)$ ovoids and spreads have q^2+1 elements.

Lemma 1.1 (an explicit spread of $W(q)$, hence an explicit ovoid of $Q(4,q)$). Choose $\alpha,\beta\in\mathbb{F}_q$ with $X^2-\beta X-\alpha$ irreducible. The q^2+1 lines

$$\ell_\infty=\langle e_2,e_3\rangle,\qquad \ell_{a,b}=\langle (1,0,a,b),\ (0,1,b,\alpha a+\beta b)\rangle\quad (a,b\in\mathbb{F}_q)$$

form a spread of $W(q)$, for every q. *Proof.* w vanishes on ℓ_∞, and $w((1,0,a,b),(0,1,b,c))=b-b=0$, so all are lines of $W(q)$. The line $\ell_{a,b}$ is $\{(u,uA):u\in\mathbb{F}_q^2\}$ with $A=\begin{pmatrix}a & b\\ b & \alpha a+\beta b\end{pmatrix}$, so $\ell_{a,b}\cap\ell_{a',b'}\neq\emptyset$ iff $A-A'$ is singular, i.e. $\delta_a(\alpha\delta_a+\beta\delta_b)-\delta_b^2=0$ with $(\delta_a,\delta_b)=(a-a',b-b')$. If $\delta_a\neq 0$, put $t=\delta_b/\delta_a$: this says $t^2-\beta t-\alpha=0$, impossible; if $\delta_a=0$ it says $\delta_b=0$. No $\ell_{a,b}$ meets ℓ_∞. So the q^2+1 lines are pairwise disjoint and cover all $(q^2+1)(q+1)$ points. □

Under the Klein correspondence these lines are q^2+1 pairwise non-collinear points of $Q(4,q)$: an ovoid O of $Q(4,q)$. (It is an elliptic quadric section, but we do not need that.)

Lemma 1.2 (facts we cite). (a) $W(q)$ has a spread for every q (Lemma 1.1); $W(q)$ has an ovoid iff q is even; dually $Q(4,q)$ has an ovoid for every q and a spread iff q is even [PT, 3.4.1, 3.4.2]. (b) $W(q)$ is self-dual iff q is even [PT, 3.2.1]; for q even $W(q)\cong Q(4,q)$. (c) All points of $W(q)$ are regular; the points of $Q(4,q)$ are regular for q even and antiregular for q odd [PT, 3.3.1]. In a GQ of order (s,s), a point x is regular iff every triad containing x has 1 or $s+1$ centres, and antiregular iff every such triad has 0 or 2 centres [PT, 1.3.6]. So for q odd: **triads of points of $Q(4,q)$ have 0 or 2 centres; triads of lines of $Q(4,q)$ (= triads of points of $W(q)$) have 1 or $q+1$ centres.** (d) (Tallini [Ta91]) A partial ovoid of $W(q)$, q odd, has at most q^2-q+1 points; equivalently a partial spread of $Q(4,q)$, q odd, has at most q^2-q+1 lines.

Computer-checked (research/theorems/d5gq/triads.py, triads_out.txt): on 3,000 random triads each, $W(q)$ shows only 1 and $q+1$ centres and $Q(4,q)$ only 0 and 2, for $q=3,5,7$.

Lemma 1.3 (an explicit partial ovoid of $W(q)$ of size $3q-1$, q odd, $q\ge 5$). (Found by squad card r2c10; research/squads/gqd5-r2/c10/.) Write points of $\mathrm{PG}(3,q)$ as (u,v) with $u,v\in\mathbb{F}_q^2$, so

that $w((u,v),(u',v')) = u\cdot v' - v\cdot u'$ with the dot product $u\cdot v = u_0v_0 + u_1v_1$. Put $v^\perp = (-v_1, v_0)$ (so $u\cdot v^\perp = 0$ iff u, v are proportional), $a_1 = (1,0)$, $a_2 = (0,1)$, $a_3 = (1,1)$, $b_i = a_{i-1}^\perp$ (indices mod 3), and

$$S = \{(a_i, c\,b_i) : i = 1,2,3,\ c \in \mathbb{F}_q^*\} \ \cup\ \{(x,0),\ (0,z^\perp)\}$$

with $x = (1,\tau)$, $\tau \notin \{0,1\}$, and $z \notin \{a_1, a_2, a_3, x\}$ (projectively). Then S is a partial ovoid of $W(q)$ with $3q-1$ points. *Proof.* For $P = (a, cb)$, $P' = (a', c'b')$: $w(P,P') = c'(a\cdot b') - c(b\cdot a')$. Same i, $c \neq c'$: $w = (c'-c)(a_i\cdot b_i) \neq 0$ since $a_i \neq a_{i-1}$. Different i, j: $a_i\cdot b_j = 0$ iff $a_i = a_{j-1}$ iff $i = j-1$, and $b_i\cdot a_j = 0$ iff $j = i-1$; for $i \neq j$ in $\mathbb{Z}_3$ exactly one of these holds, so w is a non-zero element times a non-zero element: $w \neq 0$. $(x,0)$ against (a_i, cb_i): $w = c\,(x\cdot b_i)$, zero iff $x = a_{i-1}$, excluded. $(0,z^\perp)$ against (a_i, cb_i): $w = -z^\perp\cdot a_i$, zero iff $a_i = z$, excluded. $(x,0)$ against $(0,z^\perp)$: $w = x\cdot z^\perp \neq 0$ as $z \neq x$. The points are distinct, $3(q-1)+2$ of them. □

(For $q = 3$ the point $(0,z^\perp)$ is not available, and the three orbits with $(x,0)$, $x = (1,-1)$, give 7 points, the maximum for $W(3)$; computer-checked for $q = 3,5,7,9,11,13$ with gqlib.) Computer searches find much larger partial ovoids for specific q (§10.3): 18 ($q=5$), 33 ($q=7$), 51 ($q=9$), 70 ($q=11$). *Computer-checked* (squad card r2c8, research/squads/gqd5-r2/c8/results.md): 18 is the maximum for $W(5)$ (no 19-point partial ovoid: SAT, two solvers, using the transitivity of $\mathrm{PSp}(4,5)$ on non-collinear pairs), below Tallini's bound 21. We know no explicit family of order q^2 for q odd; Ceria, De Beule, Pavese and Smaldore [CDPS] construct large partial ovoids of $W(3,q)$ for q an odd square with $3 \nmid q$ (we did not compare sizes with ours).

Lemma 1.4 (an explicit duality of $W(q)$, q even). For a line $\ell = \langle x, y\rangle$ of $W(q)$ put $\psi(\ell) = (p_{01}, p_{03}, p_{23}, p_{12})$ (Plücker coordinates as above), and for a point x let $\varphi(x)$ be the set $\{\psi(\ell) : x \in \ell\}$. For q even, ψ is a bijection from the lines to the points of $W(q)$, each $\varphi(x)$ is a line of $W(q)$, and $x \in \ell \iff \psi(\ell) \in \varphi(x)$; so (φ, ψ) is a duality. *Proof.* By §1.2, $\ell \mapsto X = (p_{01}, p_{23}, p_{02}, p_{03}, p_{12})$ maps the lines of $W(q)$ onto the points of $Q(4,q)$: $X_0X_1 + X_2^2 + X_3X_4 = 0$, and the pencil of lines through x onto a line of $Q(4,q)$. For q even the polar form $X_0Y_1 + X_1Y_0 + X_3Y_4 + X_4Y_3$ has radical $\langle e_2\rangle$ (the nucleus), and projecting from e_2, $X \mapsto (X_0, X_3, X_1, X_4) = \psi(\ell)$, maps $Q(4,q)$ bijectively onto $\mathrm{PG}(3,q)$ and lines of $Q(4,q)$ onto lines that are totally isotropic for $y_0y_2' + y_2y_0' + y_1y_3' + y_3y_1'$, which is w in characteristic 2 [PT, 3.2.1]. Composing gives the claim. □ *Computer-checked* for $q = 2, 4, 8$ (all incidences); squad card c11 built the same duality another way. The image $O_W = \psi$(spread of Lemma 1.1) is then an ovoid of $W(q)$.

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2. Level graphs

Let $\mathcal{S}$ be a GQ of order (s,s), so $|P| = |L|$; I its incidence graph. Fix $r \geq 1$ and a *regular tournament* T on the vertex set $\mathbb{Z}_{2r+1}$ ("levels"): every two levels are joined by exactly one arc, and every level has r outgoing arcs. The *rotational* tournament has the arcs $j \to j+k$, $1 \leq k \leq r$. Each level gets a type $\tau_j \in \{H, D\}$ with

$$A(H) = P, \quad B(H) = L, \qquad A(D) = L, \quad B(D) = P.$$

An arc $t = (j \to j')$ is a *transition*; it is *friendly* if $\tau_j \neq \tau_{j'}$ (then $A(\tau_j) = B(\tau_{j'})$).

Definition 2.1 (level graph). Given a bijection $M_t : A(\tau_j) \to B(\tau_{j'})$ for every arc $t = (j \to j')$, $X = X(T, \tau, M)$ has vertex set $(P \cup L) \times \mathbb{Z}_{2r+1}$ and edges $(x,j) \sim (y,j)$ for incident x, y ("level edges") and $(a,j) \sim (M_t(a), j')$ for every arc $t = (j \to j')$ and $a \in A(\tau_j)$ ("matching edges"). Colour (a,j), $a \in A(\tau_j)$, with $\mathcal{A}$ and (b,j), $b \in B(\tau_j)$, with $\mathcal{B}$. Neighbourhoods:

$$N_X(a,j) = \Gamma_1(a)\times\{j\} \cup \{(M_t(a), j') : t = (j \to j') \in T\}, \qquad N_X(b,j') = \Gamma_1(b)\times\{j'\} \cup \{(M_t^{-1}(b), j) : t = (j \to j') \in$$

X is bipartite (colours $\mathcal{A}$, $\mathcal{B}$), $(s+1+r)$-regular, of order $(2r+1)\cdot 2|P|$.

With all types H, one bijection $\sigma : P \to L$ on every arc and $(s,s) = (q,q)$ these graphs have the orders of the 2013 table values listed in §0 (two plain matchings over $\mathbb{Z}_5$, three over $\mathbb{Z}_7$, four over $\mathbb{Z}_9$).

Theorem A. For every tournament, typing and choice of bijections, X has diameter at most 5. Precisely: two vertices of one level are joined inside the level by a path of length ≤ 4; for every arc $t = (j \to j')$, every vertex of level j is joined to every vertex of level j' by a path of length ≤ 5 that uses only level edges of levels j, j' and matching edges of t.

Proof. Inside a level: (F1). Let $t = (j \to j')$, $M = M_t$, and take (u,j), (v,j'). Elements of different types are at distance ≤ 3 in I, of equal type at distance ≤ 4. 1. $u \in A(\tau_j)$: $(u,j) \to (M(u),j') \rightsquigarrow (v,j')$, length $\le 1+4$. 2. $u \in B(\tau_j)$, $v \in B(\tau_{j'})$: $(u,j) \rightsquigarrow (M^{-1}(v),j) \to (v,j')$, length $\le 3+1$. 3. $u \in B(\tau_j)$, $v \in A(\tau_{j'})$: take $b \in \Gamma_1(v)$: $(u,j) \rightsquigarrow (M^{-1}(b),j) \to (b,j') \to (v,j')$, length $\le 3+1+1$. □

Computer-checked: research/theorems/d5gq/test_basic.py (output test_basic_out.txt) built 32 level graphs with random types and random bijections over $W(q)$ and $Q(4,q)$, $q = 2,3,4,5$, $r = 1,2$; all have diameter exactly 5.

Proposition 2.2 (unequal orders). If $\mathcal{S}$ has order (s,t) with $s \neq t$ and $2r+1 \ge 3$, no typing of a tournament on the levels makes every arc friendly, and a non-friendly arc needs a bijection between P and L, which do not have the same size. So Definition 2.1 with perfect matchings exists only for $s = t$. *Proof.* Three levels span a triangle of the tournament, and two of them have the same type; the arc between them is not friendly. $|P| - |L| = (s-t)(st+1) \neq 0$. □

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3. Surgery and hubs

Definition 3.1. Start from $X(T,\tau,M)$ and choose for every arc $t = (j \to j')$ a set $R_t \subseteq A(\tau_j)$ of *freed elements*. 1. *Deletion.* Delete the matching edges $(a,j)(M_t(a),j')$, $a \in R_t$. Each leaves a free $\mathcal{A}$-slot (a,t) at (a,j) and a free $\mathcal{B}$-slot $(M_t(a),t)$ at $(M_t(a),j')$. Level j has $\alpha_j = \sum_{t \text{ out of } j} |R_t|$ free $\mathcal{A}$-slots and $\beta_j = \sum_{t \text{ into } j} |R_t|$ free $\mathcal{B}$-slots. Call the result X_R. 2. *Hubs.* Add $K_{\mathcal{B}} \le \min_j \alpha_j$ vertices of colour $\mathcal{B}$ ($\mathcal{B}$-hubs) and $K_{\mathcal{A}} \le \min_j \beta_j$ vertices of colour $\mathcal{A}$ ($\mathcal{A}$-hubs). Each $\mathcal{B}$-hub h takes one free $\mathcal{A}$-slot in every level, at the vertex $(\omega_j(h), j)$, and each $\mathcal{A}$-hub g one free $\mathcal{B}$-slot in every level, at $(\omega'_j(g), j)$; every slot is taken at most once. 3. *Hub graph.* Add a bipartite graph Y between the $\mathcal{B}$-hubs and the $\mathcal{A}$-hubs.

Call the result X_R^+. Neighbourhoods: a level vertex has the neighbourhood of Definition 2.1 minus its deleted matching edges plus one hub per taken slot; a hub has $N(h) = \{(\omega_j(h), j) : j\} \cup N_Y(h)$ and $N(g) = \{(\omega'_j(g), j) : j\} \cup N_Y(g)$. X_R^+ is bipartite (hubs join the opposite colour), level vertices have degree $\le s+1+r$, hubs $2r+1+\deg_Y$; so the maximum degree is $\le s+1+r$ as soon as Y has maximum degree $\le s-r$. The order is $(2r+1)\cdot 2|P| + K_{\mathcal{A}} + K_{\mathcal{B}}$.

(In all our constructions every hub has one neighbour in every level; Lemma 3.4 uses this. A hub cannot be joined to an $\mathcal{A}$-vertex of one level and a $\mathcal{B}$-vertex of another: all $\mathcal{A}$-vertices have one colour, so this would break bipartiteness, as squad card c17 also found by experiment.)

Lemma 3.2 (one matching edge). Let $t = (j \to j')$. A simple path of length ≤ 5 from level j to level j' that uses only level edges of j, j' and remaining matching edges of t uses exactly one

matching edge. *Proof.* It uses an odd number of them. With three or more, write the path as $(u,j) \rightsquigarrow (a_1,j) \to (M(a_1),j') \rightsquigarrow (M(a_2),j') \to (a_2,j) \rightsquigarrow (a_3,j) \to \cdots$. A simple path does not use the same edge twice, so $a_1 \neq a_2$ and $a_2 \neq a_3$; $M(a_1), M(a_2)$ have the same type, so they are at distance ≥ 2, and so are a_2, a_3. The length is then $\geq 3+2+2=7$. □

Definition 3.3. The transition t is *repaired* (by R_t) if in X_R every vertex of level j is joined to every vertex of level j' by a path of length ≤ 5 using only level edges of j, j' and remaining matching edges of t. By Lemma 3.2: iff for all u, v there is $a \in A(\tau_j) \setminus R_t$ with

$$d(u,a) + 1 + d(M_t(a), v) \ \leq \ 5. \tag{3.1}$$

Lemma 3.4 (hubs reach everything). In X_R^+ every hub is within distance 5 of every level vertex, and every $\mathcal{B}$-hub within distance 5 of every $\mathcal{A}$-hub. *Proof.* $h \to (\omega_j(h), j) \rightsquigarrow (v, j)$ has length $\leq 1+4$. For h, g: $\omega_0(h) \in A(\tau_0)$ and $\omega_0'(g) \in B(\tau_0)$ have different types, so $h \to (\omega_0(h), 0) \rightsquigarrow (\omega_0'(g), 0) \to g$ has length $\leq 1+3+1$. □

Theorem 3.5 (assembly). Suppose - (T) every transition t is repaired by R_t, and - (HG) any two $\mathcal{B}$-hubs, and any two $\mathcal{A}$-hubs, are within distance 4 in X_R^+.

Then X_R^+ has diameter ≤ 5. If moreover Y has maximum degree $\leq q - r$ (order (q,q)), then

$$n_b(q+1+r, 5) \ \geq \ (2r+1) \cdot 2N(q) + K_{\mathcal{A}} + K_{\mathcal{B}}.$$

Proof. Level pairs: (F1) inside a level, (T) between levels (every pair of levels is an arc). Pairs with a hub: Lemma 3.4 and (HG). Same-colour vertices are at even distance, so "within 5" means "within 4" for them. □

(HG) holds if Y alone has all same-colour distances ≤ 4 (§9); paths through the levels may also help.

Theorem 3.6 (assembly, general form). Theorem 3.5 remains true if (T) is replaced by - (T') for every arc $t = (j \to j')$, every vertex of level j is within distance 5 of every vertex of level j' in X_R^+.

Proof. The same; (T) implies (T') because X_R is a subgraph of X_R^+. □

We use (T') for transitions that are repaired only with the help of hubs (§7) or of other levels (§12).

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4. Theorem E: the exact criterion

Theorem E. Let $t = (j \to j')$ be any transition of a level graph over a GQ of order (s,s), $M = M_t$, $R = R_t \subseteq A(\tau_j)$, $R \neq \emptyset$. Then t is repaired if and only if - **(E2)** for every $x \in R$, every element of $A(\tau_{j'})$ is incident with $M(a)$ for some $a \in \Gamma_2(x) \setminus R$; - **(E3)** for every $x \in R$, every element of $B(\tau_j)$ is incident with some $a \notin R$ such that $M(a) \in \Gamma_2(M(x))$.

Proof. We check (3.1) for all (u,v), sorted by the colours of u (in level j) and v (in level j'). Recall: $d(u,a)$ is even iff u, a have the same type; $M(a)$ has the type of $B(\tau_{j'})$; elements of different types are at distance ≤ 3.

(i) $u \in A(\tau_j) \setminus R$: $a = u$ gives $0 + 1 + d(M(u), v) \leq 5$. Always fine.

(ii) $u = x \in R$, $v \in A(\tau_{j'})$: we need $a \neq x$ with $d(x,a) + d(M(a),v) \leq 4$; $d(x,a) \geq 2$ is even and $d(M(a),v) \geq 1$ is odd, so the only possibility is $d(x,a) = 2$, $d(M(a),v) = 1$. This holds for all v iff (E2) holds for x.

(iii) $u = x \in R$, $v \in B(\tau_{j'})$: if (E2) holds, pick $w \in \Gamma_1(v)$ (it is in $A(\tau_{j'})$) and $a \in \Gamma_2(x) \setminus R$ with $M(a)$ incident with w; then $d(x,a) + d(M(a),v) \leq 2 + 2$. So (iii) follows from (E2).

(iv) $u \in B(\tau_j)$, $v \in A(\tau_{j'})$: both distances are odd, so the possibilities are $(1,1),(1,3),(3,1)$. If some $b \in \Gamma_1(v)$ has $M^{-1}(b) \notin R$, take $a = M^{-1}(b)$: $d(u,a) \leq 3$, total ≤ 4. If some $a \in \Gamma_1(u) \setminus R$ exists, $d(M(a),v) \leq 3$ gives total ≤ 4. If neither ($\Gamma_1(v) \subseteq M(R)$ and $\Gamma_1(u) \subseteq R$), every $a \notin R$ has $d(u,a) \geq 3$ and $d(M(a),v) \geq 3$: the pair fails. So (iv) holds for all pairs iff (E1): *it is not the case that some element of $B(\tau_j)$ has all its incident elements in R and some element of $A(\tau_{j'})$ has all its incident elements in $M(R)$.* And (E2) implies (E1): if $\Gamma_1(v) \subseteq M(R)$ for some $v \in A(\tau_{j'})$, no $a \notin R$ has $M(a)$ incident with v, so (E2) fails for every $x \in R$.

(v) $u \in B(\tau_j)$, $v \in B(\tau_{j'})$: $d(u,a)$ odd, $d(M(a),v)$ even; total ≤ 4 means $(1,0),(1,2)$ or $(3,0)$. If $v \notin M(R)$, $a = M^{-1}(v)$ works $(3+0)$. If $v = M(x)$, $x \in R$: $d(M(a),v) = 0$ would force $a = x$, so we need $a \in \Gamma_1(u) \setminus R$ with $M(a) \in \Gamma_2(M(x))$; for all u this is (E3) for x.

These cases cover all pairs ($u \in A \setminus R$: (i); $u \in R$: (ii),(iii); $u \in B$: (iv),(v)). □

(If $R = \emptyset$ the transition is repaired by Theorem A.)

Computer-checked: test_basic.py compares the conditions of Theorem E (repair.py, `exact_conditions`, which also tests (E1)) with a brute-force evaluation of (3.1) over all pairs (repair.py, `brute_bad`): 30 random friendly and non-friendly transitions for each $q = 2,3,4,5$, and 300 transitions with whole lines and pencils inside R for $q = 2,3$ (transitions failing only (E2), only (E3), and both occur); they agree in every case.

---

5. Theorem B: automorphic transitions

A transition is *automorphic* if M_t is the restriction of an automorphism ψ of I with $\psi(A(\tau_j)) = B(\tau_{j'})$: a friendly transition with $M_t = \mathrm{id}$, or a non-friendly one with M_t a duality (which exists for $W(q)$ iff q is even, Lemma 1.2(b)).

Theorem B. An automorphic transition with $A(\tau_j) = P$ is repaired if and only if R_t is a partial ovoid (pairwise non-collinear). Dually, with $A(\tau_j) = L$, iff R_t is a partial spread.

Proof. Take $A(\tau_j) = P$. (E2) for x: $\psi(a)$ is incident with v iff a is incident with $\psi^{-1}(v)$, and ψ^{-1} maps $A(\tau_{j'})$ onto $B(\tau_j) = L$. So (E2) for x says: every line contains a point of $\Gamma_2(x) \setminus R$. By (F4), a line through x contains $s \geq 1$ points of $\Gamma_2(x)$ and a line ℓ not through x exactly one, p_ℓ. If $R \cap \Gamma_2(x) = \emptyset$ every line is covered. If $y \in R \cap \Gamma_2(x)$, a line $\ell \neq xy$ through y (it exists, $t \geq 1$) has $p_\ell = y \in R$ and is not covered. So (E2) for all $x \in R \iff R$ is non-collinear. (E3) for x: $M(a) \in \Gamma_2(M(x))$ iff $a \in \Gamma_2(x)$, so (E3) is the same statement. □

(A friendly transition whose matching is a collineation, restricted to points or to lines, is also automorphic; Theorem B covers it too.)

So at $D = 5$ the freed sets of automorphic transitions are exactly the partial ovoids; there is no analogue of the hexagon's segment sets (a pair of collinear freed points already breaks the transition). For $W(q)$ and $Q(4,q)$ the largest are the ovoids, $q^2 + 1$ points, when they exist (Lemma 1.2(a)).

Computer-checked: test_basic.py, 30 random sets per $q \le 5$ (half partial ovoids, half not): repaired iff partial ovoid. Squad card c9 built a three-level graph over $Q(4,5)$ with an 18-line partial spread plus one line meeting two of its lines on a friendly transition, and no hubs: the graph has diameter 6, as Theorem B predicts for the two levels.

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6. Theorem C: local dualities

For a non-friendly transition between two levels over the same quadrangle (a map $M : P \to L$ of $W(q)$ or of $Q(4,q)$) there is no duality when q is odd. We ask M to behave like one near R.

Theorem C. Let t be any transition, $M = M_t$, $R = R_t$. If - (H1) R is non-collinear (resp. non-concurrent), and - (H3) $M(\Gamma_2(x)) = \Gamma_2(M(x))$ for every $x \in R$,

then t is repaired.

Proof. (E2) for x: by (H1) $\Gamma_2(x) \cap R = \emptyset$, so $\{M(a) : a \in \Gamma_2(x) \setminus R\} = \Gamma_2(M(x))$ by (H3); every element of the other type is incident with an element of $\Gamma_2(M(x))$ by (F3). (E3) for x: $\{a \notin R : M(a) \in \Gamma_2(M(x))\} = \Gamma_2(x)$, which meets every element of $B(\tau_j)$ by (F3). □

At $D = 7$ the analogous theorem also needed (H2), "$M(R)$ non-concurrent"; at $D = 5$ it follows from (H1), (H3) anyway: for $x \ne y \in R$, $M(\{x,y\}^\perp) = \Gamma_2(M(x)) \cap \Gamma_2(M(y))$ has $s+1$ elements, which is only possible if $M(x), M(y)$ do not meet (two concurrent lines have $s-1$ common neighbours in Γ_2).

Proposition 6.1 (local dualities preserve triad centres). Under (H1), (H3), for pairwise non-collinear $x, y, z \in R$: $|\{x,y,z\}^\perp| = |\{M(x), M(y), M(z)\}^\perp|$ (the right side counted in the dual). *Proof.* $\{x,y,z\}^\perp = \Gamma_2(x) \cap \Gamma_2(y) \cap \Gamma_2(z)$, and M is a bijection with $M(\Gamma_2(\cdot)) = \Gamma_2(M(\cdot))$ on R. □

Corollary 6.2 (q odd: two points at most). Let q be odd and let $M : P \to L$ be a bijection of $Q(4,q)$ (or of $W(q)$). If R satisfies (H1) and (H3) then $|R| \le 2$. *Proof.* Three points of R are pairwise non-collinear by (H1), so they form a triad of points of $Q(4,q)$, which has 0 or 2 centres (Lemma 1.2(c)). Their images are pairwise non-concurrent (remark after Theorem C), so they form a triad of lines of $Q(4,q)$; under $Q(4,q) \cong W(q)^D$ this is a triad of points of $W(q)$ whose centres are the lines of $Q(4,q)$ meeting all three, and it has 1 or $q+1$ centres. As $q+1 \ge 4$, the counts never agree, contradicting Proposition 6.1. The same argument with the roles of $W(q)$ and $Q(4,q)$ exchanged works for $W(q)$. A $D \to D$ transition over $Q(4,q)$ is a bijection from the lines to the points of $Q(4,q)$, i.e. from the points to the lines of $W(q)$, and is covered by the $W(q)$ case. □

This is the main difference from $D = 7$, where local dualities carried 81 to 768 freed points per transition. For q even there is a global duality and Theorem B applies instead.

Proposition 6.4 (two points always work). Let x, y be non-collinear points and ℓ, m non-concurrent lines of a GQ of order (s,s). Define $M : P \to L$ by $M(x) = \ell$, $M(y) = m$, any bijection $\{x,y\}^\perp \to \{\ell,m\}^\perp$, any bijection $\Gamma_2(x) \setminus \{x,y\}^\perp \to \Gamma_2(\ell) \setminus \{\ell,m\}^\perp$, the same for y, m, and any bijection on the rest. Then (H1), (H3) hold for $R = \{x,y\}$, so the transition is repaired. *Proof.* The four sets $\{x,y\}$, $\{x,y\}^\perp$, $\Gamma_2(x) \setminus \{x,y\}^\perp$, $\Gamma_2(y) \setminus \{x,y\}^\perp$ are pairwise disjoint (as x, y are not collinear, $x \notin \Gamma_2(y)$), of sizes 2, $s+1$, s^2-1, s^2-1, and the same holds for the lines. So the pieces define a bijection between two subsets of the same size, which extends to $P \to L$; (H3) holds by construction. □

The program research/theorems/d5gq/localdual.py builds this M (the "arbitrary" bijections in sorted order) and confirms (E2), (E3) and the brute force for $W(3), Q(4,3), W(5), Q(4,5)$.

Computer-checked 6.5 (Theorem E goes beyond local dualities). Squad card c4 found, by annealing, a bijection M of $Q(4,5)$ and a freed set of **three** points that satisfies (E2), (E3) (research/squads/gqd5-r1/c4/best.json; we re-checked it with `exact_conditions` and `brute_bad`). Its triad has 2 centres and the image triad 1, so it is not a local duality. No 4-point set was found ($q = 5$), and a SAT search for 3 points in $Q(4,3)$ did not finish in 600 s.

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7. Relaxed freed sets of lines

In the typed layouts of §8 the binding constraint is the size b of the freed sets of lines on friendly transitions $D \to H$ (partial spreads of $Q(4,q)$, i.e. partial ovoids of $W(q)$). Here is how to free more lines.

Fix a friendly transition $t = (j \to j')$ with $\tau_j = D$, $\tau_{j'} = H$, $M_t = \mathrm{id}$, over $Q(4,q)$ (so $A(\tau_j) = L$, $B(\tau_j) = P$, $A(\tau_{j'}) = P$, $B(\tau_{j'}) = L$). Let $T = R_t$ be a set of lines in which every line meets at most one other line of T, its *partner*. For a partner pair ℓ, m let $z = \ell \cap m$.

Proposition 7.1 (the bad pairs of a partner pair). Inside levels j, j', the pairs at distance > 5 are exactly: for every partner pair (ℓ, m), the pairs $((\ell, j), (p, j'))$ and $((p, j), (\ell, j'))$ with $p \in m \setminus \{z\}$ (and the same with ℓ, m exchanged): $4q$ ordered pairs per partner pair. *Proof.* We go through the cases of the proof of Theorem E with $M = \mathrm{id}$ and $R = T$. Case (i) is always fine. Case (iv): a point v of level j' lies on $q + 1 \geq 3$ lines, at most two of them in T, so some line through v is not in $M(T) = T$. Lines of T without partner: $\Gamma_2(\ell) \cap T = \emptyset$, and cases (ii), (iii), (v) hold as in Theorem B. Now let ℓ have partner m, $z = \ell \cap m$. Case (ii), $u = (\ell, j)$ and v a point of level j': a point $v \notin \ell$ lies on exactly one line $\lambda(v)$ meeting ℓ ((F4) dually), and v is incident with an element of $\Gamma_2(\ell) \setminus T$ iff $\lambda(v) \neq m$ (the only line of T meeting ℓ), i.e. iff $v \notin m \setminus \{z\}$; a point $v \in \ell$ lies on $q \geq 2$ lines of $\Gamma_2(\ell)$, at most one of them in T. So exactly the pairs $((\ell, j), (v, j'))$, $v \in m \setminus \{z\}$, fail. Case (iii), $u = (\ell, j)$ and v a line of level j': v has a point $w \notin m \setminus \{z\}$ (if $v \neq m$, v meets m in at most one point; if $v = m$ take $w = z$). By case (ii) some $a \in \Gamma_2(\ell) \setminus T$ is incident with w, and $d(\ell, a) + d(a, v) \leq 2 + 2$: fine. Case (v), $v = (\ell, j')$ and u a point of level j: we need a line $a \notin T$ through u meeting ℓ; by the same count this exists iff $u \notin m \setminus \{z\}$. So exactly the pairs $((u, j), (\ell, j'))$, $u \in m \setminus \{z\}$, fail. □

Proposition 7.2 (the hub rule). Suppose that for every line $\ell \in T$ with partner m, the $\mathcal{A}$-hub g_ℓ that takes the $\mathcal{B}$-slot (ℓ, t) (at the vertex (ℓ, j')) has its slot in level j at a point of m: $\omega'_j(g_\ell) \in m$. Then every pair of Proposition 7.1 is at distance ≤ 4 in X_R^+.

Proof. Let $p \in m \setminus \{z\}$. The pair $((p, j), (\ell, j'))$: $(\ell, j') \to g_\ell \to (\omega'_j(g_\ell), j) \to (m, j) \to (p, j)$, a path of length ≤ 4 (shorter if $\omega'_j(g_\ell) = p$). The pair $((\ell, j), (p, j'))$: the partner of m is ℓ, so the hub g_m of the slot (m, t) sits at (m, j') and, in level j, at a point $\omega'' = \omega'_j(g_m)$ of ℓ; hence $(\ell, j) \to (\omega'', j) \to g_m \to (m, j') \to (p, j')$ has length 4. □

So a transition with a relaxed set is not repaired in the sense of Definition 3.3, but with the hub rule it satisfies (T') of Theorem 3.6. (An earlier version of this note also asked the $\mathcal{B}$-hub of the slot (ℓ, t) to sit, in level j', on m; the second path above shows that this is not needed. We found this when a graph violating that condition for 128 slots passed the all-pairs check.)

The hubs exist. In our layouts the natural slot on m is the $\mathcal{B}$-slot at the point o_m where the ovoid O meets m (O meets every line exactly once), freed on an $H \to D$ transition into level j. One $\mathcal{A}$-hub

can serve the line ℓ on all transitions out of level j at once, because they all ask for a point of m in level j; so the rule needs one $\mathcal{A}$-hub per partnered line and per D-level, at most $|T| \le K_{\mathcal{A}}$ of them. Distinct partnered lines ask for distinct points: if $o_m = o_{m'}$ for $m \neq m'$, then m and m' meet in a point of O, so they are partners with meeting point in O. We therefore require $z \notin O$ for every partner pair. The $\mathcal{B}$-hubs are not constrained at all.

In the language of $W(q)$ (lines of $Q(4,q)$ = points of $W(q)$, the ovoid O = the spread of Lemma 1.1), T is a set of points of $W(q)$ in which every point is collinear with at most one other, and the line joining a collinear pair is not in the spread. Counting lines (a single point uses $q+1$ lines, a pair $2q+1$, no line is shared) gives $|T| \le 2(q+1)(q^2+1)/(2q+1)$, i.e. at most 86 for $q=9$ and 127 for $q=11$. *Computer-checked*: tabu searches find such sets of size 66 for $q=9$ (squad card r2c1; 30 pairs) and 87 for $q=11$ (card r2c2), against 51 and 70 for partial ovoids. For $q=5$ the maximum is 23 (exhaustive search with the group, card r2c8), against the counting bound 28 and the maximum partial ovoid 18. Squad card r2c7 tested the rule independently at $q=5$ (five levels $HHDHD$ over $Q(4,5)$, 23 freed lines with 6 partner pairs, $25+25$ hubs): with the hub rule the graph is valid (1,610 vertices, apsp), with the same sets and a hub assignment ignoring the rule it has diameter 6 (research/squads/gqd5-r2/c7/). Card r3c4 tested the final form (only the $\mathcal{A}$-hub condition, $\mathcal{B}$-hubs placed arbitrarily, many partner pairs): five levels over $Q(4,5)$ with 23 lines and 6 pairs (1,610 vertices) and over $Q(4,7)$ with 41 lines and 15 pairs (4,086 vertices) are valid, and moving the level-j slot of one $\mathcal{A}$-hub off the partner line gives diameter 6 in both cases (research/squads/gqd5-r3/c4/).

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8. Typed layouts and slot counts

By Theorem 3.5 the order is $(2r+1)\cdot 2N + K_{\mathcal{A}} + K_{\mathcal{B}}$ with $K_{\mathcal{B}} \le \min_j \alpha_j$ and $K_{\mathcal{A}} \le \min_j \beta_j$. Over $Q(4,q)$, q odd, write a for the size of the freed sets on $H \to D$ transitions (points of $Q(4,q)$: an ovoid, $a = q^2+1$), b for $D \to H$ (lines of $Q(4,q)$), c for non-friendly transitions ($c=2$ by Proposition 6.4, $c \le 2$ for local dualities by Corollary 6.2).

Below $a \ge b \ge c \ge 0$.

Proposition 8.1 (seven levels). For the rotational tournament on $\mathbb{Z}_7$ ($j \to j+1, j+2, j+3$) with types $HHHDHHD$:

$$\alpha = (a{+}2c,\ a{+}2c,\ a{+}2c,\ 2b{+}c,\ a{+}2c,\ a{+}2c,\ 3b), \qquad \beta = (b{+}2c,\ b{+}2c,\ b{+}2c,\ 3a,\ b{+}2c,\ b{+}2c,\ 2a{+}c).$$

So $\min\alpha = \min(a+2c, 2b+c, 3b)$ and $\min\beta = b+2c$ (as $b+c \le 2a$); for $a+c \le 2b$, $K_{\mathcal{A}} + K_{\mathcal{B}} \le a+b+4c$. *Proof.* Direct count: level 0 (H) has arcs to 1 (H), 2 (H), 3 (D), so $\alpha_0 = c+c+a$; level 3 (D) to 4, 5 (H), 6 (D): $b+b+c$; level 6 (D) to 0, 1, 2 (H): $3b$; and similarly for the others and for incoming arcs. □

Computer-checked (research/theorems/d5gq/layouts.py; squad card c15, research/squads/gqd5-r1/c15/formulas.md): over all 2,640 labelled regular tournaments on 7 vertices and all typings, the largest $\min\alpha + \min\beta$ is $\min(a+b+4c,\ 3b+3c)$ for $a \ge b \ge c \ge 0$; for five levels it is $2b+2c$ (types $HHDHD$, Proposition 8.2 below); for nine levels $4b+4c$ (exhaustive over the 15 isomorphism classes); for 11 and 13 levels sampling gives $\min(2a+2b+6c, 5b+5c)$ and $6b+6c$ (lower bounds only).

Proposition 8.2 (five levels). Rotational tournament on $\mathbb{Z}_5$, types $HHDHD$: $\alpha = (a+c, a+c, b+c, a+c, 2b)$, $\beta = (b+c, b+c, 2a, b+c, a+c)$, so $\min\alpha = \min\beta = b+c$. (Direct count.)

Proposition 8.3 (nine levels). Rotational tournament on $\mathbb{Z}_9$ ($j \to j+1, \dots, j+4$), types $HHHHDHHHD$:

$\alpha = (a{+}3c,\ a{+}3c,\ a{+}3c,\ a{+}3c,\ 3b{+}c,\ a{+}3c,\ a{+}3c,\ a{+}3c,\ 4b)$, $\beta = (b{+}3c,\ b{+}3c,\ b{+}3c,\ b{+}3c,\ 4a,\ b{+}3c,\ b{+}3c,$

so $\min\alpha = \min(a+3c, 3b+c)$ and $\min\beta = b+3c$. (Direct count; also computed by layouts.py.)

With three levels some level has no friendly outgoing arc, so $K_{\mathcal{A}}, K_{\mathcal{B}} \le c$.

For q even we use instead all levels H over $W(q)$, $M_t = \varphi|_P$ for a duality φ on every arc and $R_t = O_W$ an ovoid of $W(q)$ on every arc: every transition is repaired by Theorem B and $\alpha_j = \beta_j = r(q^2+1)$.

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9. Hub graphs

Write $\mathrm{HG}_5(K_{\mathcal{B}}, K_{\mathcal{A}}; \delta)$ for a bipartite graph with colour classes of sizes $K_{\mathcal{B}}$, $K_{\mathcal{A}}$, maximum degree $\le \delta$, in which same-colour vertices are within distance 4. If Y is such a graph with $\delta \le q - r$, both conditions of Theorem 3.5 on Y hold.

Lemma 9.1 (explicit hub graphs). $\mathrm{HG}_5(K_{\mathcal{B}}, K_{\mathcal{A}}; \delta)$ exists in each of the following cases (s a prime power): - (Y0) $1 \le K_{\mathcal{B}}, K_{\mathcal{A}} \le \delta$ (complete bipartite graph); - (Y1) $s+1 \le \delta$ and $s^2{+}s{+}1 \le K_{\mathcal{B}}, K_{\mathcal{A}} \le (s^2{+}s{+}1)(\delta{-}s)$; - (Y2) $s{+}1 \le \delta$ and $N(s) \le K_{\mathcal{B}}, K_{\mathcal{A}} \le N(s){+}(s{+}1)(\delta{-}s{-}1)$; - (Y3) $K_{\mathcal{B}} = K_{\mathcal{A}} = (2r'+1)N(s)$ with $s+1+r' \le \delta$.

Proof. (Y0): distances 2. (Y1): the incidence graph of $\mathrm{PG}(2,s)$ (points as $\mathcal{B}$, lines as $\mathcal{A}$; diameter 3, same-colour distance 2) with pendant vertices: up to $\delta - s - 1$ new $\mathcal{B}$-vertices on each line and new $\mathcal{A}$-vertices on each point. A pendant p on c and a core vertex x of the colour of p: c and x have different colours, $d(c,x) \le 3$, so $d(p,x) \le 4$. Two pendants of one colour hang on core vertices of one colour, at distance ≤ 2: $d \le 4$. (Y2): the incidence graph of $W(s)$ (same-colour distance ≤ 4, fact (F1)) with up to $\delta - s - 1$ pendant $\mathcal{B}$-vertices on each of the $s+1$ lines through a fixed point z_0 and pendant $\mathcal{A}$-vertices on each of the $s+1$ points of a fixed line m_0: a pendant on a line c and a core point are within $1+3$; two pendants on lines through z_0 within $1+2+1$; dually. (Y3): the level graph of Theorem A over $W(s)$ with $2r'+1$ levels has degree $s+1+r'$ and diameter 5, so same-colour distance ≤ 4. □

Squad card c10 (research/squads/gqd5-r1/c10/table.md) tabulated the largest explicit balanced sizes (5, 15, 45, 120, 255, 468, 780, 1200, 2000, 2925, 4100 for $\delta = 2, \dots, 12$; for $\delta = 3$ from (Y2), for $\delta \ge 4$ from (Y3), and for $\delta = 2$ the 10-cycle, which is not in Lemma 9.1), and found that random δ-regular bipartite graphs satisfy the condition up to about 271 vertices per colour at $\delta = 8$. Squad card r2c9 (research/squads/gqd5-r2/c9/construction.md, gen_hubs.py) extends (Y1), (Y2) by joining several $\mathrm{PG}(2,s)$ cores: every balanced size $7 \le K \le 48$ at $\delta = 6$ and $7 \le K \le 153$ at $\delta = 8$ is realised (some members found by a seeded search; each checked by BFS) (*Computer-checked*). For the records we use a computer-found Y (random degrees, then double-edge swaps that minimise the number of same-colour hub pairs at distance > 4 in X_R^+, paths through the levels counted; research/theorems/d5gq/hubopt.py), and check the whole graph.

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10. Families valid for every q

All ingredients below are explicit: the spread of Lemma 1.1 (an ovoid of $Q(4,q)$), the partial ovoid of Lemma 1.3, the local dualities of Proposition 6.4, the hub graphs of Lemma 9.1, and for q even

a duality of $W(q)$.

Corollary 10.1 (q even, $1 \le r \le q-1$). Let all levels have type H over $W(q)$, $M_t = \varphi|_P$ for the duality φ of Lemma 1.4, and $R_t = O_W$ the ovoid of Lemma 1.4 on every arc. Then for any $K_{\mathcal{B}}, K_{\mathcal{A}} \le r(q^2+1)$ for which an $\mathrm{HG}_5(K_{\mathcal{B}}, K_{\mathcal{A}}; q-r)$ exists (Lemma 9.1),

$$n_b(q+1+r, 5) \;\ge\; (2r+1)\cdot 2N(q) + K_{\mathcal{A}} + K_{\mathcal{B}}.$$

Proof. Theorem B repairs every transition; Theorem 3.5 with Y the hub graph. □

Corollary 10.2 (q odd, $q \ge 5$, five levels). Types $HHDHD$ over $Q(4,q)$ (Proposition 8.2) with the ovoid O on $H \to D$, the lines of $Q(4,q)$ corresponding to the points of Lemma 1.3 on $D \to H$ ($b = 3q-1$, $q \ge 5$), Proposition 6.4 on the non-friendly arcs: for $K_{\mathcal{A}}, K_{\mathcal{B}} \le 3q+1$ with an $\mathrm{HG}_5(K_{\mathcal{B}}, K_{\mathcal{A}}; q-2)$,

$$n_b(q+3, 5) \;\ge\; 10N(q) + K_{\mathcal{A}} + K_{\mathcal{B}}.$$

Corollary 10.3 (q odd, $q \ge 5$, seven levels). Types $HHHDHHD$ (Proposition 8.1), same ingredients: for $K_{\mathcal{B}} \le \min(q^2+5,\, 6q)$, $K_{\mathcal{A}} \le 3q+3$ with an $\mathrm{HG}_5(K_{\mathcal{B}}, K_{\mathcal{A}}; q-3)$,

$$n_b(q+4, 5) \;\ge\; 14N(q) + K_{\mathcal{A}} + K_{\mathcal{B}}.$$

Corollary 10.4 (q odd, $q \ge 5$, nine levels). Types $HHHHDHHHD$ (Proposition 8.3), same ingredients: for $K_{\mathcal{B}} \le \min(q^2+7,\, 9q-1)$, $K_{\mathcal{A}} \le 3q+5$ with an $\mathrm{HG}_5(K_{\mathcal{B}}, K_{\mathcal{A}}; q-4)$,

$$n_b(q+5, 5) \;\ge\; 18N(q) + K_{\mathcal{A}} + K_{\mathcal{B}}.$$

Proofs. Theorems B (friendly arcs: the ovoid is a partial ovoid, the set of Lemma 1.3 gives a partial spread of $Q(4,q)$) and C/Proposition 6.4 (non-friendly arcs) repair every transition; slot counts from Propositions 8.1–8.3 with $a = q^2+1$, $b = 3q-1$, $c = 2$; Theorem 3.5. □

10.3 With computer-found sets. Replacing b by the size of a computer-found partial ovoid of $W(q)$ (Lemma 1.2(d) bounds it by q^2-q+1) or a relaxed set (§7), and the hub graph by a computer-found one, gives the larger values of §14. The best partial ovoids of $W(q)$ we found: 18 ($q=5$), 33 ($q=7$), 51 ($q=9$; four ILS runs and a group-orbit search all stop at 50–51), 70 ($q=11$; tabu search; 68 from unions of orbits).

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11. The $(13,5)$ record graph

Computer-checked 11.1. $n_b(13,5) \ge 11{,}636$.

Construction ($q=9$, $r=3$; built by research/theorems/d5gq/wgr.py with no extra freed points, i.e. the layout of build_relaxed.py; solution file research/theorems/d5gq/graphs/d13D5_q9_relaxed66_n11636.jso
edges …n11636.edges.gz). - $G = Q(4,9)$ as `W(9).dual()` (§1.2); levels $\mathbb{Z}_7$, rotational tournament, types $HHHDHHD$. - $H \to D$ arcs ($0\to3$, $1\to3$, $2\to3$, $4\to6$, $5\to6$): $M = \mathrm{id}$, $R = O$, the ovoid of Lemma 1.1 (82 points). - $D \to H$ arcs ($3\to4$, $3\to5$, $6\to0$, $6\to1$, $6\to2$): $M = \mathrm{id}$, $R = T$, a relaxed set of 66 lines with 30 partner pairs (§7), listed in the solution file (`T`; also graphs/W9_relaxed_set_66.json as points of $W(9)$; found by squad card r2c1). - The eleven non-friendly arcs ($0\to1$, $0\to2$, $1\to2$, $1\to4$, $2\to4$, $2\to5$, $4\to5$, $4\to0$, $5\to0$, $5\to1$ of type $H\to H$, and $3\to6$ of type $D\to D$): $M =$ the local duality of Proposition 6.4 on a two-element

freed set, given in full in the solution file (`nonfriendly`). - Slots: $\alpha = (86, 86, 86, 134, 86, 86, 198)$, $\beta = (70, 70, 70, 246, 70, 70, 166)$; $K_{\mathcal{B}} = 86$, $K_{\mathcal{A}} = 70$ hubs; the slot of every hub in every level is listed (`Bhub_slots`, `Ahub_slots`, as triples (level, element, arc)). The hub rule of Proposition 7.2 holds at all $5 \cdot 60 = 300$ (arc, partnered line) instances (research/theorems/d5gq/check_rule.py). - Y: 420 edges (`Y`), maximum degree 5 on the $\mathcal{B}$-side and $6 = q - r$ on the $\mathcal{A}$-side.

Every two vertices in different levels are within distance 5: by Theorems B and C for the $H \to D$ and the non-friendly arcs, and by Propositions 7.1 and 7.2 for the $D \to H$ arcs; (HG) holds (checked over the whole graph by the builder); so Theorem 3.6 gives order $7 \cdot 2 \cdot 820 + 86 + 70 = 11{,}636$, maximum degree 13. **Computer-checked**: research/lifts/coordinator_check/apsp.c (BFS from every vertex) reports bipartite, degree max 13, diameter 5 (graphs/d13D5_q9_relaxed66_n11636.check.txt). The neighbourhood of every vertex is determined by the solution file through Definitions 2.1 and 3.1. The solution file also lists the 820 points of $\mathrm{PG}(3, 9)$ and the 820 lines of $W(9)$ explicitly; research/theorems/d5gq/sol2edges.py re-checks that they form $W(9)$, rebuilds the graph from the file alone (without the builder's code) and finds exactly the same edge set.

Earlier versions: 11,605 and 11,621 with a 51-point partial ovoid instead of the relaxed set (without and with the local dualities; graphs/d13D5_q9_typed_n11621.json); 11,626 and 11,627 with relaxed sets of 56 and 57 lines and at most 14 pairs, built with the stronger hub rule (graphs/d13D5_q9_relaxed_n11627.json).

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12. Routing through a third level (computer experiments)

Theorem E is about two levels. With $2r + 1 \geq 5$ levels a pair $(u, j), (v, j')$ can also be joined through a third level i with two matching edges, e.g. $(x, j) \to (M_{j \to i}(x), i) \to (z, i) \to (M_{i \to j'}(z), j') \to (v, j')$ (length 4). *Computer-checked* (squad cards): - c19: seven levels over $Q(4, 5)$, all type H: a **random** 16-point freed set on one non-friendly arc, violating (E2) and (E3), becomes harmless after annealing the other 20 matchings (apsp VALID, 2,184 vertices); five levels: 8 points. - c6: three levels over $Q(4, 3)$: 5 freed points per transition with hubs (10 hubs, valid), all violating (E2) or (E3). - r2c5: the typed seven-level layout of Proposition 8.1 over $Q(4, 5)$ with c freed points on **every** $H \to H$ arc, hubs included, annealing matchings, hub slots and Y: valid for $c = 2, 3, 4, 5$ (2,236, 2,240, 2,244, 2,248 vertices; research/squads/gqd5-r2/c5/). - r2c6: seven levels over $Q(4, 7)$, no hubs: 16 freed points on one arc, 8 on each of two arcs, 4 on each of four arcs (all valid). - The director's annealer research/theorems/d5gq/wgr.py (targeted swaps in the non-friendly matchings and moves of hub slots; bad pairs counted by wgr_eval.c on the whole graph with hubs but without Y, which is chosen at the end): at $q = 5$ it reaches zero bad pairs with 4 freed points per $H \to H$ arc (the hub graph then fails (HG) because $q - r = 2$). At $q = 9$, on top of the 11,636 graph of §11 with one extra freed point on each of the ten $H \to H$ arcs (which would give 11,640), three runs went from about 2,650 bad pairs to 10–16 in 200,000–250,000 swaps and then stalled; a greedy polish with 150 candidate moves per step did not close the last ones. (A first version also allowed swaps in the $D \to D$ arc $3 \to 6$; pairs between levels 3 and 6 can then not be repaired at all, because every route through a third level uses only friendly arcs. The later runs keep that arc fixed.)

Heuristic (why this does not scale to $q = 11$): for a freed x and a random M, about N/e elements of the other level violate (E2). A third level supplies about $(q + 1)^2$ length-4 routes per freed element and per route type, so the fraction of $A(\tau_{j'})$ left uncovered after the five other levels is about $e^{-1}e^{-5(q+1)^2/N}$: 12% for $q = 5$ but 22% for $q = 11$, i.e. about 330 pairs per freed element to repair by deliberate changes, each change repairing $q + 1$ of them. We did not get third-level

routing to work at $q \geq 9$.

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13. Quadrangles of order (s, t) with $s \neq t$

Proposition 2.2 rules out the level graph for $Q(5, q)$, $H(3, q^2)$ and $AS(q)$. Small experiments (squad cards c12, c13; research/squads/gqd5-r1/c12, c13) tried the natural substitutes at the cells $\Delta = 5, 7, 9$: - $GQ(2, 4)$ at $\Delta = 5$: the 27 points already have degree 5, so only the 45 lines (2 spare edges each) can carry matchings, and the level structure must be bipartite (no odd cycles of levels). Best valid graph: 144 vertices (two levels), against the lab's 354. - $AS(q)$ at $\Delta = q + 2$, in the model "copies of the incidence graph of $AS(q)$ plus line-to-line edges, no other vertices": points have degree $q + 2$ and lines degree q, so only lines have spare edges (two each), and line-to-line edges must join copies of opposite types. From a point p of one copy, a point of another copy of the same type is at even distance, and a path of length 4 to it must be p, a line through p, a line of a copy of the other type, a line of the target copy, a point on it; so p reaches at most $2(q + 2)q$ such points, fewer than the q^3 points of one copy when $q \geq 5$ (*Theorem*, by this count). Hence for $q \geq 5$ at most one copy of each type can be used, and the order stays $\leq 4q^2(q + 1)$. We found no construction from a quadrangle of unequal order that approaches the lab's values at $\Delta \leq 9$.

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14. Comparison with the 2013 table and the lab's values

"Theorem family" = Corollaries 10.1–10.4 (explicit sets and hub graphs; research/theorems/d5gq/family_table.py, family_table_out.txt); each of these six values was also built and checked by all-pairs BFS with independent scripts (research/theorems/d5gq/family_check.py, family_check_even.py, written by the referee agent; output family_check_out.txt). "Computer" = the best graph of §§8–11 type that we built and checked by all-pairs BFS. "Lab" = the lab's value before this note (README, research/RESULTS.md).

cell	table (2013)	theorem family	computer (this note)	lab before	construction of the lab value
(10,5)	4,000	4,044 ($q = 7$, $r = 2$)	4,062 ($q = 7$, $r = 2$)	**4,800**	doubled-matching lift of $W(7)$
(11,5)	5,850	6,090 ($q = 8$, $r = 2$)	—	**7,020**	doubled-matching lift of $W(8)$
(12,5)	8,200	8,430 ($q = 8$, $r = 3$)	—	**9,840**	doubled-matching lift of $W(9)$
(13,5)	11,480	11,552 ($q = 9$, $r = 3$)	**11,636**	11,486	—
(14,5)	14,760	14,812 ($q = 9$, $r = 4$)	—	**17,568**	doubled-matching lift of $W(11)$

cell	table (2013)	theorem family	computer (this note)	lab before	construction of the lab value
(15,5)	20,496	20,598 ($q = 11$, $r = 3$)	20,713 ($q = 11$, $r = 3$)	**20,736**	route-chart lift
(16,5)	27,300	26,453 ($q = 11$, $r = 4$)	—	**33,792**	route-chart lift

At $\Delta \leq 9$ (cells $(5,5)$–$(9,5)$) the level graphs have $6N(q)$ or $10N(q)$ vertices for small q (e.g. $(9,5)$: $6N(7) = 2{,}400$, $(7,5)$: $6N(5) = 936$) and the hubs add only $O(q)$; they are below the 2013 table there, and far below the lab's values.

Why we could not reach $(15,5)$ (*Heuristic*; nothing here is proved). With seven levels over $Q(4,11)$, $K_{\mathcal{B}} \leq \min(a + 2c, 2b + c, 3b)$ with $a = 122$, and $K_{\mathcal{A}} \leq b + 2c$, where b is the size of the freed sets on the $D \to H$ arcs. The lab's value needs $K_{\mathcal{A}} + K_{\mathcal{B}} \geq 241$, i.e. $b \geq 111$ when $c = 2$. For partial ovoids of $W(11)$ this is exactly Tallini's bound $q^2 - q + 1$ (best found: 70). Relaxed sets (§7) are not limited by Tallini's theorem, and Proposition 7.2 puts no limit on their number of pairs; counting allows up to 127 points, but the best we found has 87 (squad card r2c2; longer tabu runs plateau at 84–87), which gives 20,713 (built and checked, §15). At $q = 5$ the true maximum (23) is 82% of the counting bound; the same ratio at $q = 11$ would give about 104, still below 111. So we would need relaxed sets of $W(11)$ about a quarter larger than any we found, or several more freed points on every non-friendly arc (each one on each of the ten $H \to H$ arcs adds 4 vertices). Third-level routing gives such points at $q = 5$ and 7 (§12), but we did not get it to work at $q = 11$ in this session.

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15. Computer checks

All graphs were checked with research/lifts/coordinator_check/apsp.c (all-pairs bit-parallel BFS, no symmetry): simple, bipartite, maximum degree, exact diameter.

graph	file (research/theorems/d5gq/)	result
$(13,5)$, $n = 11{,}636$	graphs/d13D5_q9_relaxed66_n11636.g6	VALID, degree 13, diameter 5; independent rebuild (sol2edges.py) gives the same edges; hub rule checked (check_rule.py)
$(13,5)$, $n = 11{,}627$	graphs/d13D5_q9_relaxed_n11627.g6	VALID; independent rebuild identical
$(13,5)$, $n = 11{,}621$	graphs/d13D5_q9_typed_n11621.json (json; rebuild with build_typed.py)	VALID
$(15,5)$-type, $q = 11$, $n = 20{,}713$	graphs/d15D5type_q11_relaxed87_n20713.g6 (relaxed set of 87 lines; rebuild with sol2edges.py)	VALID (beats the table, not the lab)

graph	file (research/theorems/d5gq/)	result
$(15,5)$-type, $q=11$, $n=20{,}696$	build_typed.py 11 3 HHHDHHD with the 70-point set	VALID
$(10,5)$-type, $q=7$, $r=2$, $n=4{,}062$	build_typed.py 7 2 HHDHD	VALID
$(8,5)$-type, $q=5$, $r=2$, $n=1{,}596$ and $1{,}600$	build_typed.py 5 2 HHDHD (and card c7 independently)	VALID
$(8,5)$-type, $q=5$, relaxed line sets, $n=1{,}610$	squad card r2c7 (research/squads/gqd5-r2/c7/ld23.edges)	VALID; without the hub rule diameter 6
relaxed sets, A-hub rule only, $q=5,7$, $n=1{,}610$ and $4{,}086$	squad card r3c4 (research/squads/gqd5-r3/c4/)	VALID; rule broken for one line: diameter 6
explicit families (Cor. 10.1–10.4): $(10,5)$ 4,044, $(11,5)$ 6,090, $(12,5)$ 8,430, $(13,5)$ 11,552, $(14,5)$ 14,812, $(15,5)$ 20,598	family_check.py, family_check_even.py (family_check_out.txt)	all VALID
32 level graphs, $q \le 5$	test_basic.py	all diameter 5 (Theorem A)
Theorem E vs brute force triads	test_basic.py triads.py	agree on 720 transitions
		$0/2$ in $Q(4,q)$, $1/(q+1)$ in $W(q)$, $q=3,5,7$
local dualities	localdual.py	repaired, $q=3,5$
duality of $W(q)$, $q=2,4,8$	squad card c11 (research/squads/gqd5-r1/c11/)	verified; all-H $q=4$, $r=1$: 544 VALID

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16. Prior art

- **The base graphs.** The values of the cells $(10,5)$–$(15,5)$ of the 2013 table [MS13] are the orders of the level graphs of Theorem A. We could not open the CombinatoricsWiki page to see whom it credits for these $D=5$ cells (it returned an error on 26 Oct 2026); at $D=7$ the analogous hexagon cells are credited to Delorme, Gómez and Quisquater ("On large bipartite graphs", submitted, not found), and we assume the same origin here. Compounds of a bipartite graph with a tournament go back to Gómez and Fiol [GF85]. Delorme [De85] §3 builds cyclic multi-level graphs from a duality of a quadrangle, and §7–8 duplicates sets of vertices of a quadrangle's incidence graph (an ovoid/spread copy at $D=4$). We claim none of these.
- **Hubs and deleted matching edges.** R. Rajiv (Zenodo, 30 Sept 2026, doi 10.5281/zenodo.23052995) deletes the edges of a type-preserving matching at a set of pairwise non-collinear points of a compound of generalized-polygon incidence graphs and attaches new vertices joined by a graph of diameter $\le n$, for general (non-bipartite) graphs of diameter $n=4,6$. Our friendly transitions with Theorem B are the bipartite, diameter $n+1$ version of that step, and our hubs are one-sided. The lab's literature check (research/lit/novelty-2026-10-25.md, branch claude/novelty-literature-check-4i5frp) found nothing on: an exact criterion like Theorem E, local dualities (Theorem C) and their two-point limit for q odd (Corollary 6.2), relaxed

freed sets with the hub rule (§7). That check was done for the hexagon case; we did not repeat it for quadrangles, so "nothing found" applies only to those sources.

- **The lab's earlier $D = 5$ work.** The doubled-matching lifts of $W(q)$ (research/d10D7/DM_D5.md) give $12N(q)$ at degree $q + 3$ and are better than everything here for $\Delta = 10, 11, 12, 14$. The previous $(13, 5)$ value 11,486 was $7 \cdot 2N(9) + 6$ (research/cell147/).
- **Finite geometry used.** Payne–Thas [PT] for all facts on quadrangles; Tallini [Ta91] for partial ovoids of $W(q)$, q odd; Ceria–De Beule–Pavese–Smaldore [CDPS] for large partial ovoids of $W(3, q)$. The explicit family of Lemma 1.3 was found by squad card r2c10; we did not find it in these sources.

References. [CDPS] M. Ceria, J. De Beule, F. Pavese, V. Smaldore, *On large partial ovoids of symplectic and Hermitian polar spaces*, J. Combin. Des. 31 (2023) 5–22, arXiv:2203.04553. [De85] C. Delorme, *Grands graphes de degré et diamètre donnés*, European J. Combin. 6 (1985) 291–302. [GF85] J. Gómez, M. A. Fiol, *Dense compound graphs*, Ars Combin. 20-A (1985) 211–237. [MS13] M. Miller, J. Širáň, *Moore graphs and beyond: a survey of the degree/diameter problem*, Electron. J. Combin. DS14v2 (2013). [PT] S. E. Payne, J. A. Thas, *Finite Generalized Quadrangles*, 2nd ed., EMS, 2009. [Ta91] G. Tallini, *Blocking sets with respect to planes in PG(3,q) and maximal spreads of a nonsingular quadric in PG(4,q)*, Mitt. Math. Sem. Giessen 201 (1991) 141–147.